

EE2080: LECTURE 1 (31 July 2026)

PLAN: 1) Random Vectors: Defn, Mean, Correlation, Covariance, Examples.

2) Random Processes: Defn, Examples.

Finite Dimensional Distributions: Consistency

Properties: Mean, Autocorrelation, Autocovariance

REF: 1) Bruce Hajek, "Random Processes for Engineers"

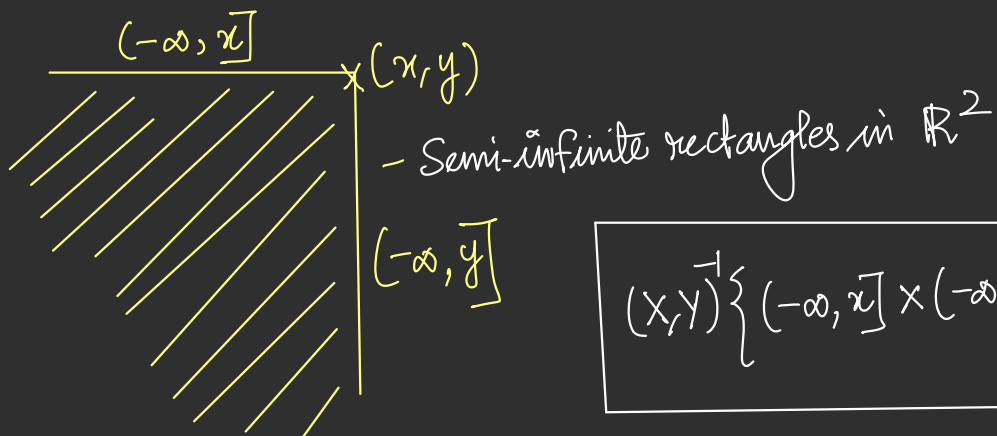
2) Anurag Kumar, "Discrete Event Stochastic Processes"

Random Vectors:

- Bivariate Random Vector: Fix a probability space $(\Omega, \mathcal{F}, P)$

Defn: Given two functions $X: \Omega \rightarrow \mathbb{R}$ and $Y: \Omega \rightarrow \mathbb{R}$,
we say $(X, Y): \Omega \rightarrow \mathbb{R}^2$ is a **bivariate random vector** w.r.t $\mathcal{F}$ if

$$\{\omega: X(\omega) \leq x\} \cap \{\omega: Y(\omega) \leq y\} \in \mathcal{F} \quad \forall x, y \in \mathbb{R}$$



$$(X, Y)^{-1} \{ (-\infty, x] \times (-\infty, y] \} \in \mathcal{F} \quad \forall x, y \in \mathbb{R}$$

Since this must hold $\forall x, y \in \mathbb{R}$

$$\Rightarrow \underbrace{\{\omega: X(\omega) \leq x\} \cap \{\omega: Y(\omega) < \infty\}}_{\Omega} \in \mathcal{F} \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \{\omega: X(\omega) \leq x\} \in \mathcal{F}, \quad \forall x \in \mathbb{R} \quad \Rightarrow X \text{ is a random variable.}$$

Similarly, Y is a random variable.

(X, Y) is a random vector $\Rightarrow X, Y$ are random variables.

Qn: Is the reverse implication true?

i.e., If X and Y are random variables w.r.t $\mathcal{F}$, is (X, Y) a random vector w.r.t $\mathcal{F}$?

Answer: YES!!

Proof: $A, B \in \mathcal{F} \Rightarrow A^c, B^c \in \mathcal{F} \Rightarrow (A^c \cup B^c) \in \mathcal{F}$
 $\Rightarrow (A^c \cup B^c)^c \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$

$\therefore X$ is a RV w.r.t $\mathcal{F} \Rightarrow \{\omega: X(\omega) \leq x\} \in \mathcal{F} \quad \forall x \in \mathbb{R}.$

Y is a RV w.r.t $\mathcal{F} \Rightarrow \{\omega: Y(\omega) \leq y\} \in \mathcal{F} \quad \forall y \in \mathbb{R}.$

Both RVs are defined w.r.t the same $(\Omega, \mathcal{F}, \mathbb{P})$

$\therefore \{\omega: X(\omega) \leq x\} \cap \{\omega: Y(\omega) \leq y\} \in \mathcal{F} \quad \forall x, y \in \mathbb{R}.$

$\Rightarrow (X, Y)$ is a random vector w.r.t $\mathcal{F}$

(X, Y) is a random vector $\Leftrightarrow X, Y$ are random variables.

Going forward, random vectors will be defined as vectors of random variables, all defined on the same probability space!!

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A random vector X w.r.t $\mathcal{F}$ of dimension m has the form $X = [X_1 \ X_2 \ \dots \ X_m]^T$, where X_i s are random variables defined w.r.t $\mathcal{F}$.

Expectation of X : $\mathbb{E}[X] = [\mathbb{E}[X_1] \ \mathbb{E}[X_2] \ \dots \ \mathbb{E}[X_m]]^T$

Auto Correlation of X : $E[XX^T]$ — $m \times m$ correlation matrix.

$$XX^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \dots & x_m \end{bmatrix} = \begin{bmatrix} x_1^2 & x_1 x_2 & \dots & x_1 x_m \\ x_2 x_1 & x_2^2 & \dots & x_2 x_m \\ \vdots & \vdots & \ddots & \vdots \\ x_m x_1 & \dots & \dots & x_m^2 \end{bmatrix}$$

$$E[XX^T]_{i,j} = E[x_i x_j]$$

Auto Covariance of X : $Cov(X)$

$$\begin{aligned} Cov(X)_{i,j} &= Cov(x_i, x_j) = E[(x_i - E(x_i))(x_j - E(x_j))] \\ &= E[x_i x_j] - E[x_i]E[x_j] \end{aligned}$$

Cross Correlation and Cross Covariance :

Consider two random vectors X of dimension m and Y of dimension n defined w.r.t the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

The cross-correlation between X and Y is the $m \times n$ matrix $E[XY^T]$ with

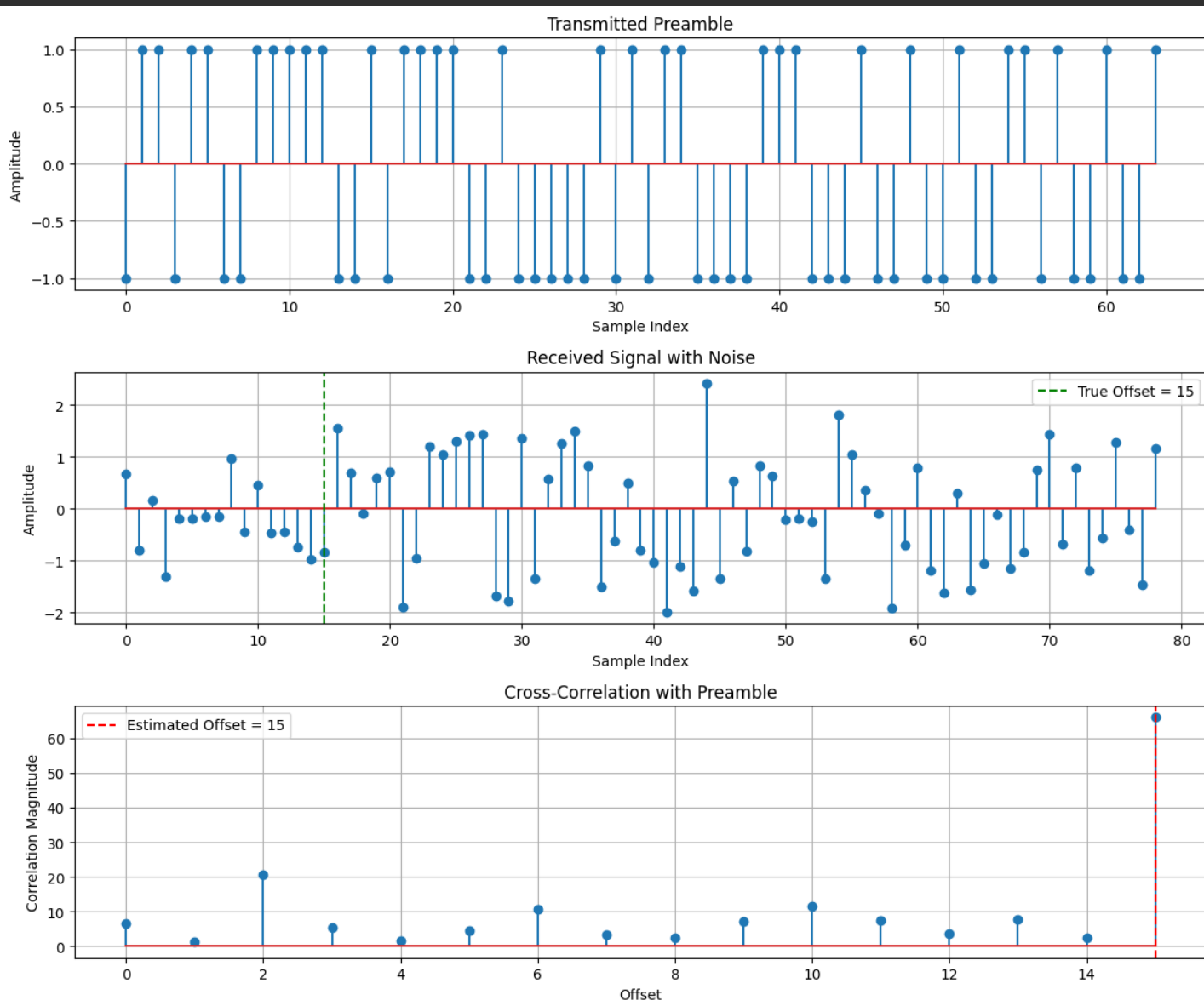
$$E[XY^T]_{i,j} = E[x_i y_j]$$

The cross-covariance between X and Y is the $m \times n$ matrix $Cov(X, Y)$ given by

$$Cov(X, Y) = E[(X - E[X])(Y - E[Y])^T]$$

$$Cov(X, Y)_{i,j} = Cov(x_i, y_j).$$

Applications of Cross-Correlation



- 1) Timing Synchronization
 - 2) Signal Detection
 - 3) Channel Estimation
- In Communication
& Signal Processing

In Computer Vision:

Similarity between two images or image regions. In tasks such as object detection, image alignment, feature/template matching.

A pattern is slid across the given image and similarity scores are computed using cross correlation. For example: Identifying a face in a group photo:

Example 1: $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}\right)$

$$Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = AX + b, \quad A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \text{ and } b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Compute $E[Y]$, $\text{cov}(Y)$ and $\text{cov}(X, Y)$.

$$E[Y] = AE[X] + b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{cov}(Y) = A \text{cov}(X) A^T = \begin{bmatrix} 8.2 & 2.8 \\ 2.8 & 1 \end{bmatrix}$$

$$\text{cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)^T] = \begin{bmatrix} 2.6 & 0.8 \\ 2.8 & 1 \end{bmatrix}$$

Example 2: $X_1 \sim \text{unif}[0, 1]$, $X_2 = X_1 + W$, $W \sim \text{unif}[0, 1]$, $X_1 \perp W$

Let $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Find $E[X]$, $E[XX^T]$ and $\text{cov}(X)$. (Homework)

Example 3: Consider $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, where x_1, x_2 are random variables defined on the same probability space with joint distribution

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 6x_2, & 0 \leq x_1 \leq 1, 0 \leq x_2 \leq x_1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$ and $E[XX^T]$

(Homework)

RANDOM PROCESS : Definition

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A random process is a collection of random variables $\{X_t, t \in \mathcal{T}\}$, indexed by the elements of an index set $\mathcal{T}$, all defined w.r.t $\mathcal{F}$.

- When $\mathcal{T}$ is finite, say $\mathcal{T} = \{1, 2, \dots, d\}$, we have a random vector $X = [X_1 \dots X_d]^T$
- If $\mathcal{T}$ is a countably infinite set, $\{X_t, t \in \mathcal{T}\}$ is a discrete parameter process
- If $\mathcal{T}$ is uncountably infinite $\{X_t, t \in \mathcal{T}\}$ is a continuous parameter process.
- Often, $\mathcal{T}$ has the interpretation of time.
- Random processes are also called Stochastic Processes (from a Greek word for "guess")

Interpretations of a Random Process $\{X_t, t \in \mathcal{T}\}$

$X_t : \Omega \rightarrow \mathbb{R}$ is a random variable for each $t \in \mathcal{T}$

$X(\omega) : \mathcal{T} \rightarrow \mathbb{R}$ is a sample path of the process for each $\omega \in \Omega$.

$X : \mathcal{T} \times \Omega \rightarrow \mathbb{R}$ $X_t(\omega)$ or $X(t, \omega)$ is a real number for each $\omega \in \Omega, t \in \mathcal{T}$.
as a function of two variables.

